WBR Perception Implementation

Understanding the system and its current evidence

Intel RealSense D435i

Raspberry Pi 5 / ROS 2 Jazzy

ORB-SLAM3 stereo-inertial

STM32H7 controller boundary

Figure 1 identifies the conceptual platform.



Figure 1: Platform objects; no validation claim.

1. Navigation and balance

KNOWN ownership / integration PLANNED

- A wheel-biped robot (WBR) needs scene-relative motion for navigation.
- The Raspberry Pi estimates motion; STM32H7 maintains balance locally.
- Perception supplies optional navigation information.

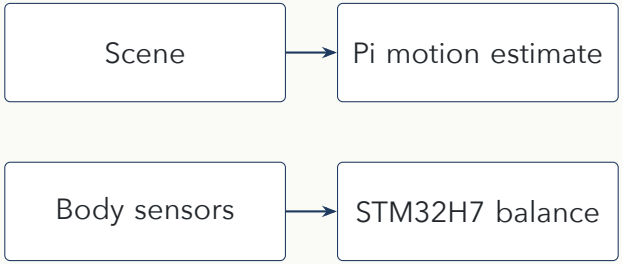


Figure 2: Navigation and balance use separate sensing paths.

Loss of perception must leave local balance control running. (Figure 2)

2. Controller ownership

KNOWN ownership / integration PLANNED

- Linux runs image processing and mapping.
- STM32H7 owns body estimation, balance and motor safety.
- Navigation references cross only a reviewed boundary.

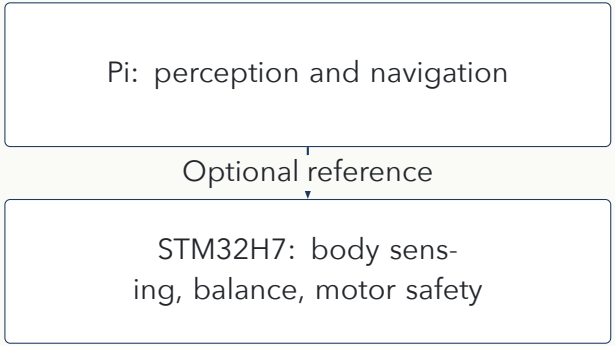


Figure 3: The controller retains local authority.

A pose is advice to navigation, never a motor-safety authority. (Figure 3)

3. Pose usability

KNOWN method

- A pose is an estimated position and orientation.
- Correct-looking numbers may describe old motion or the wrong frame.
- Consumers need age, frame and estimator validity.

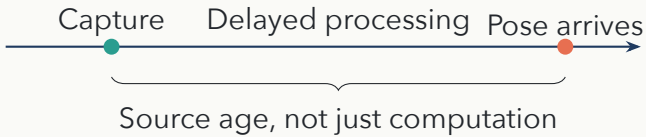


Figure 4: Delay can make a plausible pose unusable.

A recent heartbeat does not establish recent sensor capture. (Figure 4)

4. Session hardware

MEASURED boot / final availability UNVERIFIED

- The 8 GB Pi booted the same 32 GB micro Secure Digital (microSD) card.
- It reported 7.8 GiB memory; under-voltage persisted.
- The 64 GB Universal Serial Bus (USB) stick cannot sustain recording.

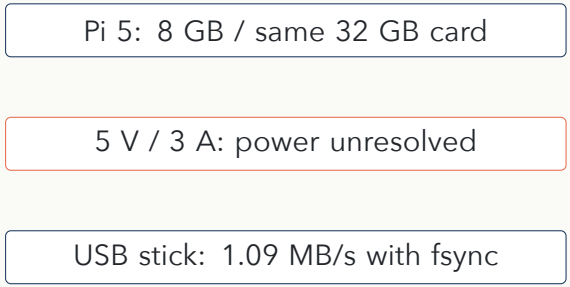


Figure 5: More memory does not repair power or storage.

The ordered 5 A supply and larger card have no qualification evidence. (Figure 5)

5. D435i sensor inputs

MEASURED identity / diagnostic development profile

- Infrared (IR) left and right images provide stereo views.
- The gyroscope measures angular rate; the accelerometer measures specific force.
- This new camera needs its own calibration and gate evidence.

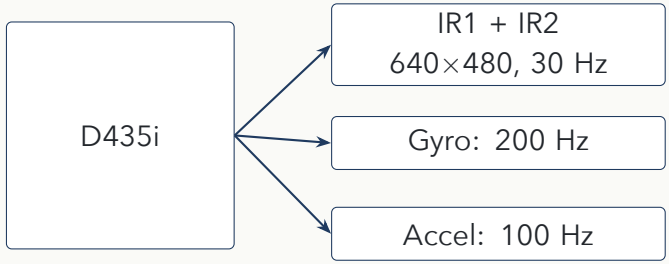


Figure 6: Selected development inputs; the emitter is off.

This paired 30 Hz profile does not replace N2's IR1/15 profile. (Figure 6)

6. Independent inertial sensors

KNOWN method

- An inertial measurement unit (IMU) combines gyro and acceleration measurements.
- The camera IMU supports visual-inertial estimation.
- The controller body IMU supports local balance.

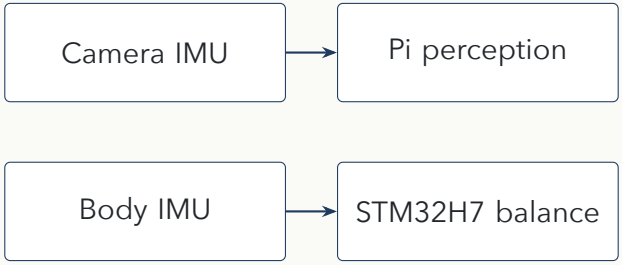


Figure 7: Different mounts, clocks and calibration; separate jobs.

Sharing an IMU name does not make the measurements interchangeable. (Figure 7)

7. Sensor-to-trajectory paths

BUILT direct path / ROS integration PLANNED

- Programs use the librealsense software development kit (SDK) directly.
- One recording can be replayed repeatedly; live mode bypasses the file.
- Robot Operating System 2 (ROS 2) integration is planned.

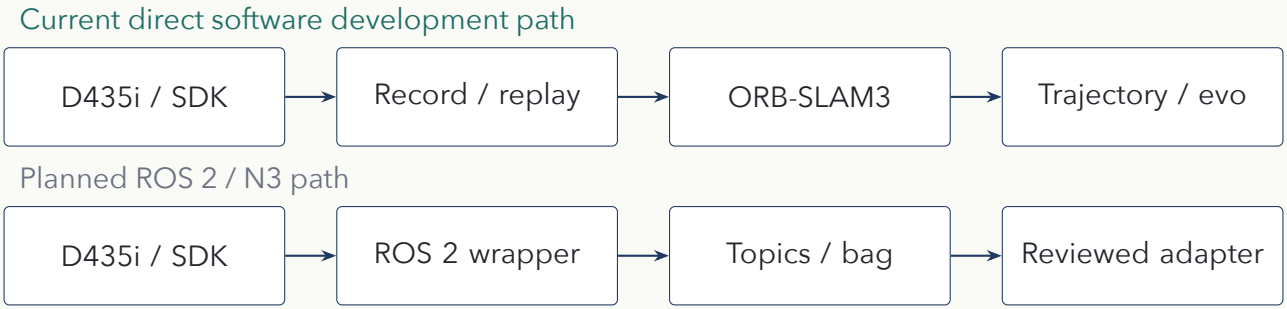


Figure 8: Current and planned paths are different implementation stages.

No successful D435i trajectory is established yet. (Figure 8)

KNOWN ROS concepts / integration PLANNED

- A publisher sends typed messages on a topic; subscribers receive them.
- Images, IMU samples and poses travel on separate topics.
- Quality of Service (QoS) policies control delivery compatibility.

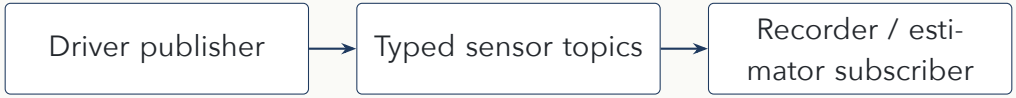


Figure 9: ROS 2 connects processes through a message contract.

Matching topic names alone does not establish compatible delivery. (Figure 9)

9. Pose origin and axes

KNOWN method

- Simultaneous Localization and Mapping (SLAM) estimates pose while building a map.
- The exported trajectory describes camera IMU/body pose in the SLAM world.
- Robot-body pose also needs the mount transform.



$$T_{WB} = T_{WI} T_{IB}$$

T_{IB}: separately defined rigid mount

Figure 10: Compose the estimated sensor pose with a defined mount.

A camera trajectory is not automatically a robot-body trajectory. (Figure 10)

KNOWN method

- Match the same physical point in the two images.
- Horizontal displacement is disparity d.
- With focal length f and baseline b, depth is $Z = fb/d$.

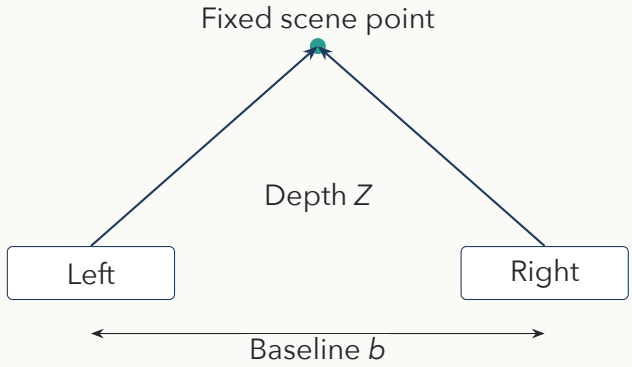


Figure 11: Two viewpoints constrain metric depth.

Bad correspondences or calibration yield bad depth. (Figure 11)

11. Features constrain motion

KNOWN method

- An image feature is a recognizable corner or local pattern.
- Match fixed scene features across stereo views and later images.
- Many consistent matches constrain camera movement.

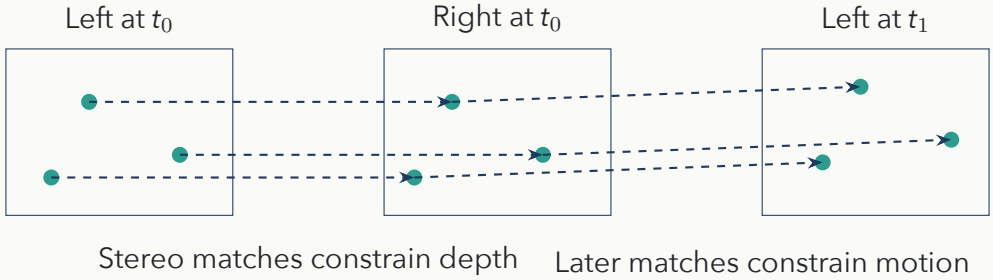


Figure 12: Correspondences link observed pixels to one motion model.

Track scene features, then test whether their geometry agrees. (Figure 12)

KNOWN method

- Visual-inertial odometry (VIO) combines both sensor cues.
- IMU prediction bridges image intervals but accumulates drift.
- Image constraints correct that prediction against the scene.

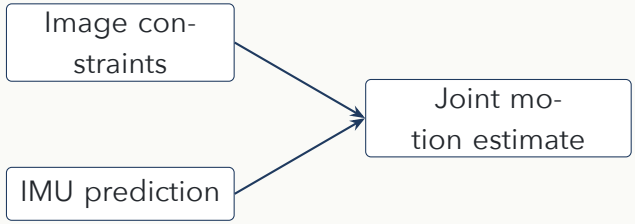


Figure 13: Two complementary cues constrain the same motion.

Calibration and timing must connect both measurements correctly. (Figure 13)

PREPARED helper / full runtime association UNVERIFIED

- Gyro and accel samples arrive between successive stereo pairs.
- The helper interpolates accel only within valid gyro-time brackets.
- Preintegration summarizes interval motion before visual correction.

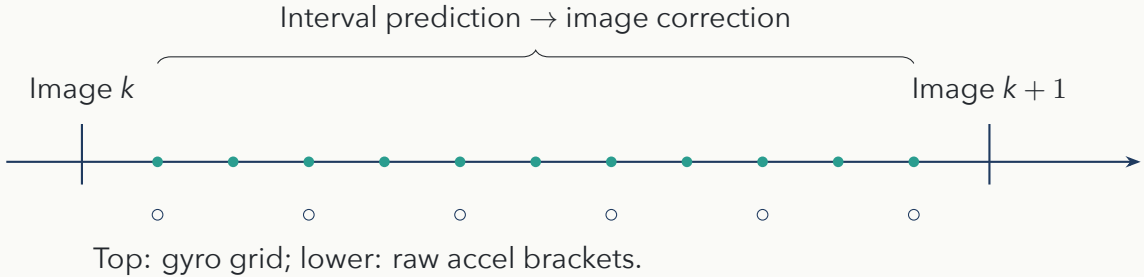


Figure 14: Interpolation aligns samples; it creates no new sensor information.

Reject missing coverage and invalid time steps rather than extrapolating. (Figure 14)

KNOWN method

- Extract and match features from each stereo pair.
- Use the inertial prediction to guide pose estimation.
- Check consistency; update tracking state and select keyframes.

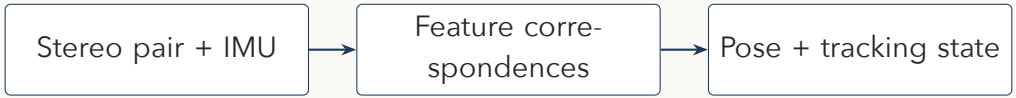


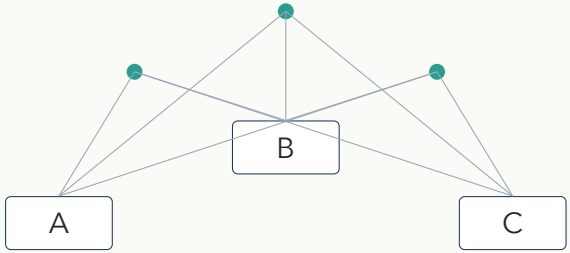
Figure 15: Tracking estimates the current sensor pose.

Tracking state is part of the output's meaning. (Figure 15)

15. Keyframes and Atlas

KNOWN method

- A keyframe preserves a useful viewpoint instead of every image.
- Mapping refines keyframe poses and persistent scene points.
- Atlas holds multiple maps; a new map is not seamless tracking.



Keyframes A–C observe shared map points.

Figure 16: Selected views preserve geometric constraints.

Inspect resets and map identity as well as trajectory files. (Figure 16)

16. Algorithm work sharing

KNOWN method

- Tracking handles each incoming image and predicts current pose.
- Local mapping refines nearby keyframes and scene structure.
- Loop closing recognizes revisits and corrects map drift.

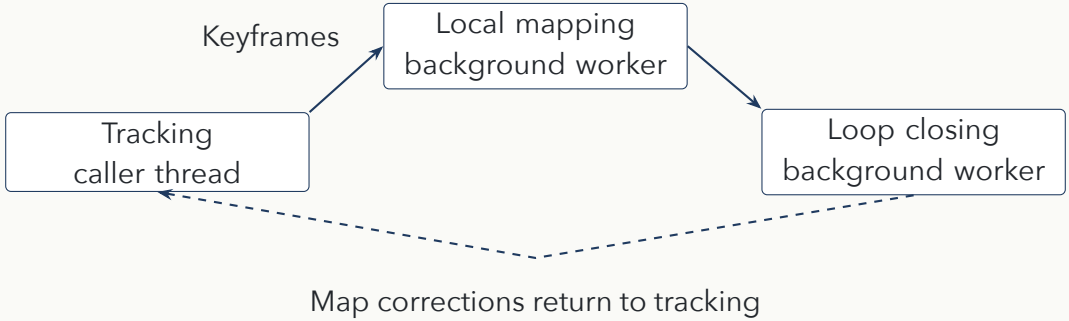


Figure 17: Roles cooperate asynchronously; queueing and scheduling matter.

Offline playback does not eliminate multithreaded run variation. (Figure 17)

17. Drift and revisits

KNOWN method

- Odometry accumulates local motion errors along a path.
- Loop closure recognizes a previously observed place.
- A map correction can change the global trajectory.

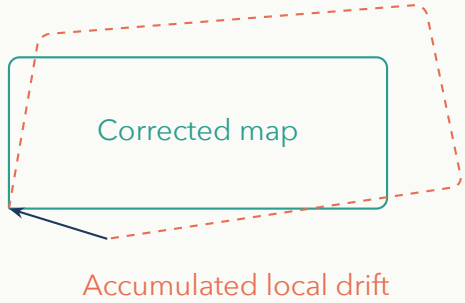


Figure 18: Conceptual paths, not measured trajectories.

Local continuity and global map correction need different frame semantics. (Figure 18)

18. Inertial initialization motion

PLANNED diagnostic recipe / not target calibration

- Initialization estimates gravity, velocity and inertial biases.
- Translate in several directions and rotate the camera.
- Then move through a small loop and finish briefly still.

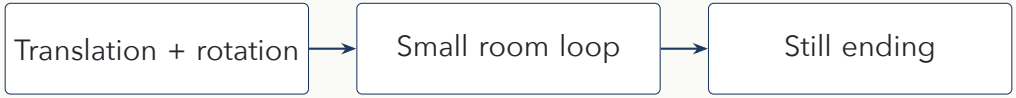


Figure 19: Diagnostic motion gives the estimator useful excitation.

Look for IMU initialization; a launch alone is not success. (Figure 19)

19. Infrared projector

DEVELOPMENT profile: emitter OFF

- Tracking needs features belonging to the fixed scene.
- Projected dots depend on the moving camera projector.
- This development path disables the emitter and uses scene texture.

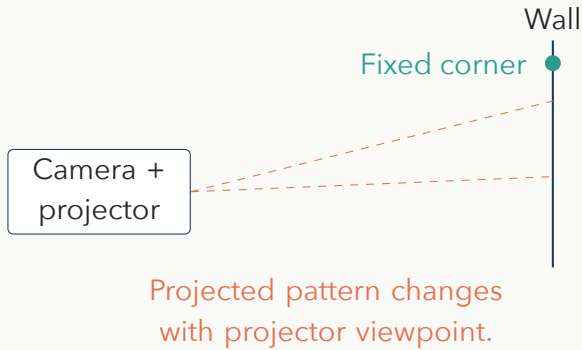


Figure 20: Scene geometry and projected texture are different cues.

Emitter off is a runtime choice; it is not calibration. (Figure 20)

20. Calibration questions

Calibration unverified

- Image intrinsics describe projection; stereo extrinsics relate the cameras.
- Camera–IMU geometry is factory nominal; physical alignment is unverified.
- Noise is assumed and time offset remains unknown.

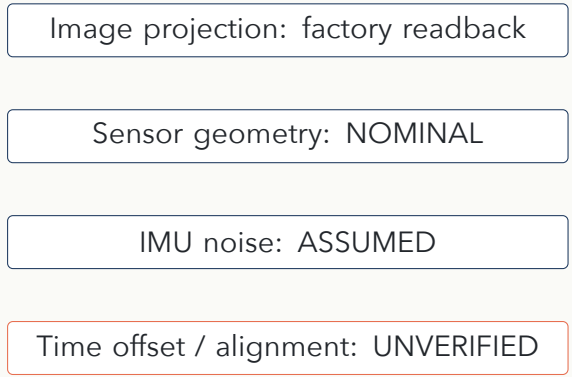


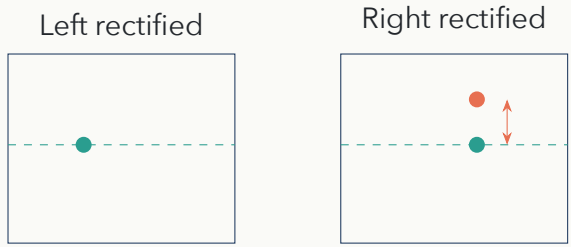
Figure 21: Four calibration questions need separate evidence.

A runnable settings file does not establish calibrated VIO. (Figure 21)

21. Stereo rectification

KNOWN method

- Rectification places corresponding points along the same row.
- A wrong image model leaves vertical mismatch or distorted geometry.
- Validate paired pixels before tuning the estimator.



Vertical mismatch indicates a geometry problem.

Figure 22: Correct matches share an epipolar row.

Matched rates do not establish matched image geometry. (Figure 22)

22. Three different capture motions

PLANNED qualification / diagnostic recipe available

- Target calibration estimates geometry and camera–IMU timing.
- Room motion tests diagnostic initialization and tracking.
- A stationary noise recording supports Allan analysis.

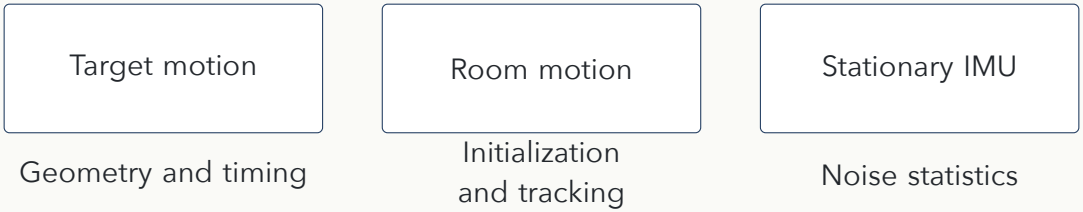


Figure 23: Different captures answer different engineering questions.

A successful hand-motion demo does not replace calibration. (Figure 23)

23. Timestamp clock domains

KNOWN method

- Sensor stamps describe capture in a device clock.
- Host receipt and processing times use a host clock.
- Shared units do not establish shared epochs or drift.

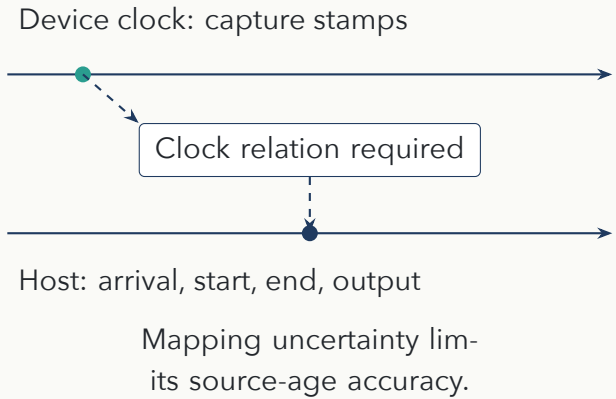


Figure 24: Transport links events, not automatically their clock origins.

Never subtract device capture from host output without a defined relation. (Figure 24)

24. Queues and backpressure

KNOWN method

- A slow consumer lets old samples accumulate in a queue.
- Blocking may reach acquisition and cause recording gaps.
- Bound queues and make overflow or rejection visible.



Oldest leaves first; sustained arrival exceeds consumption.

Figure 25: Queue direction follows arrival order and dequeue order.

Low compute time can coexist with high source age. (Figure 25)

25. Recording gaps on the card

MEASURED gaps / cause INFERRED

- The 100 s hand-moved take had 0.3–2.9 s gaps after 22 s.
- Observed file growth was 16.6 MB/s against an 18.4 MB/s input budget.
- Writer stalls blocking acquisition are strongly inferred.

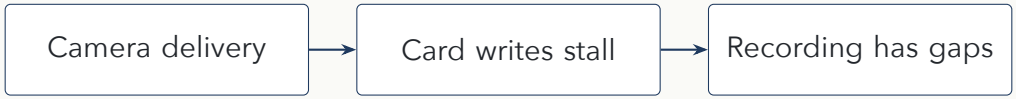


Figure 26: Observed gaps; storage bottleneck is an inferred cause.

No USB/power kernel error during this take does not resolve general power. (Figure 26)

26. Record in RAM, copy later

MEASURED 15 s diagnostic / long take PLANNED

- Random-access memory (RAM) avoids card writes during capture.
- The 15 s test: IR 29.99 Hz, maximum gap 33 ms, zero observed drops.
- Gyro 200.2 Hz; accel 101 Hz. The later copy passed checksum.



Figure 27: Short RAM test separated capture from slow persistent writes.

3.9 GB RAM filesystem fits an estimated 1.7 GB take; long reliability is unverified. (Figure 27)

27. Camera-independent baseline

MEASURED historical 4 GB baseline / N4A OPEN

- EuRoC MH01 is a public drone dataset with stereo and IMU measurements.
- Ground truth (GT) is independently measured motion used as the reference answer.
- The Pi completed three trials with one map, no resets and IMU initialization.

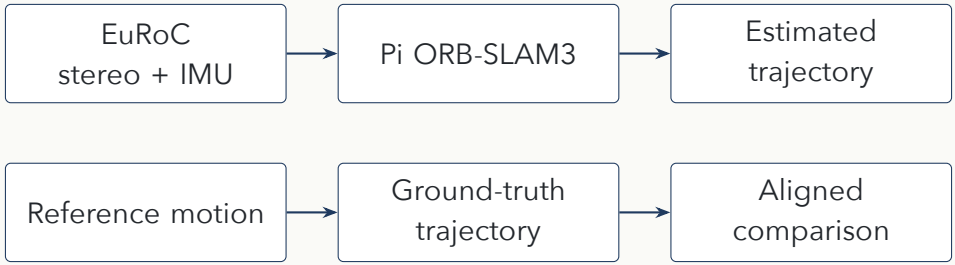


Figure 28: Known input separates algorithm/build issues from camera adaptation.

These completed trials do not establish D435i accuracy or real-time speed. (Figure 28)

28. Baseline trajectory error

MEASURED ATE / N4A qualification OPEN

- Absolute Trajectory Error (ATE) is position mismatch after aligning the paths.
- Use one rotation and translation, with no scale fitting (SE(3)).
- Root mean square error (RMSE) summarizes the position residuals.

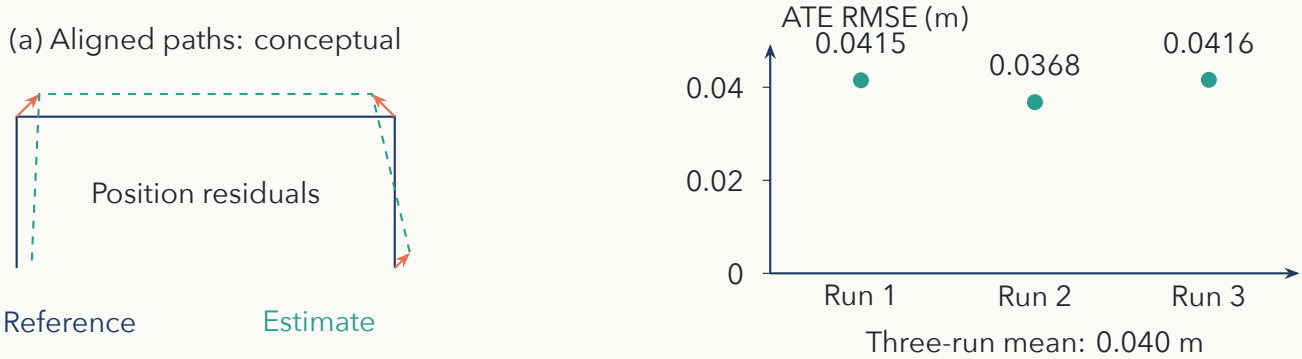


Figure 29: (a) Residuals after rigid alignment; (b) measured MH01 ATE, no scale fitting.

About four centimetres on MH01; D435i accuracy remains UNKNOWN. (Figure 29)

29. Baseline success and camera crash

MEASURED crash / rebuild INCOMPLETE

- Three initial compatibility patches enabled the native build and EuRoC trials.
- Offline D435i replay crashed before the first keyframe.
- Patch 4 initializes the keyframe pointer; rebuilding was interrupted by power loss.

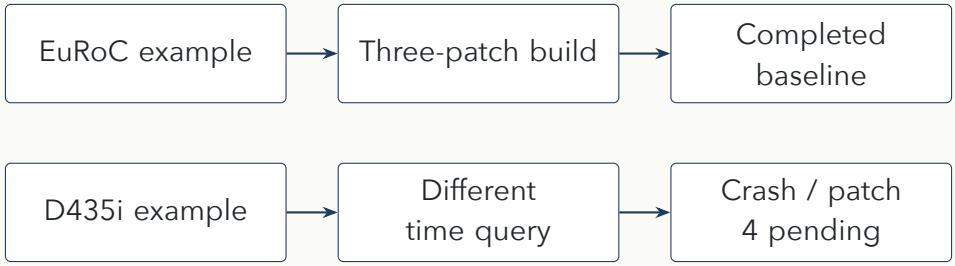


Figure 30: The two examples exercise different code paths.

A source repair is not a verified executable until rebuilding succeeds. (Figure 30)

30. Repeatable offline diagnosis

MEASURED recording / successful N4B UNVERIFIED

- The recorder preserves camera samples and timestamps in one file.
- Replay runs the same input without a live acquisition deadline.
- The first offline smoke test crashed; no camera trajectory succeeded.

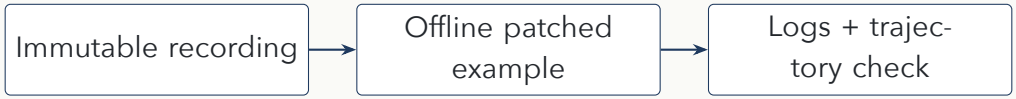


Figure 31: Repeated input isolates estimator changes from acquisition changes.

Resume and verify patch-4 rebuilding before another diagnostic run. (Figure 31)

31. Live freshness

PLANNED live diagnostic / N4C OPEN

- Live estimation must consume new data as it arrives.
- Queue growth can make an otherwise accurate pose stale.
- No successful live estimator run is documented; rebuild is incomplete.



Figure 32: A conceptual overload timeline; no latency measurement.

Measure age, queueing and drops after offline diagnosis succeeds. (Figure 32)

32. Bottom-up diagnosis

KNOWN method

- Identify the earliest failed layer before tuning later layers.
- A recording gap starts with delivery/storage checks.
- A first-frame crash starts with its stack trace and build identity.

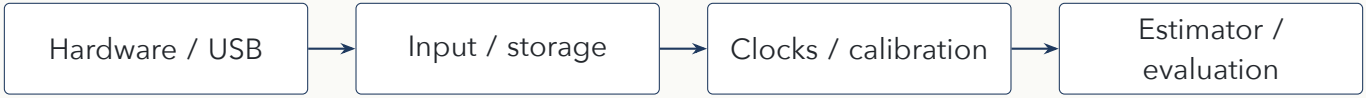


Figure 33: Earlier failures invalidate conclusions from later layers.

Preserve raw warnings and failed runs while isolating the cause. (Figure 33)

33. Health supervision

Integration PLANNED / no physical safety verdict

- A heartbeat says a process is alive, not that its pose is fresh.
- A planned receiver checks source age, validity and session identity.
- Stale or invalid perception withdraws optional navigation input.

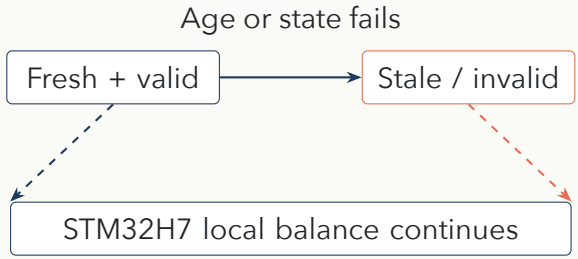


Figure 34: Navigation validity is supervised separately from balance.

Rates, age budgets and message design still require review. (Figure 34)

34. Different evidence branches

MEASURED diagnostics / all gates OPEN

N4A tests known data; camera qualification follows N0 → N1 → N2 → N3. No gate has PASSED.

Exact N2 raw profile: IR1 Y8 640×480/15, gyro/200, accel/100.

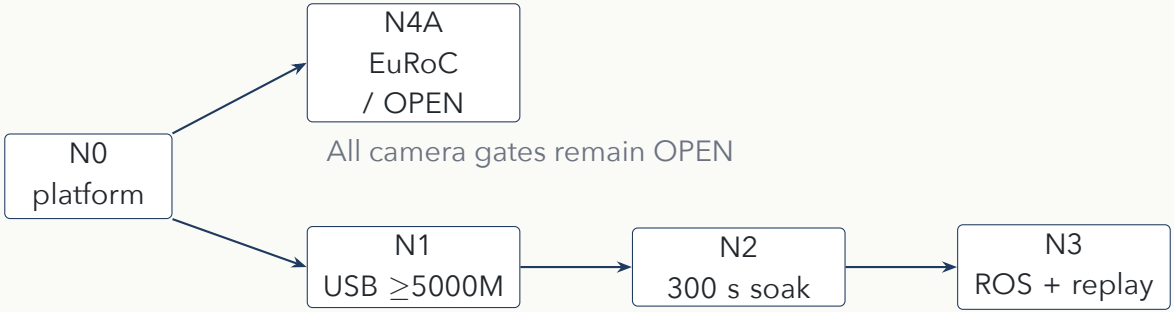


Figure 35: Dataset results do not qualify new-camera transport or timing.

The 15 s RAM test cannot replace the mandatory 300-second N2 soak. (Figure 35)

35. Established work and open gaps

MEASURED progress / power UNRESOLVED

- Established: 8 GB boot, EuRoC trials and short RAM recording.
- Open: power, long capture, patch-4 rebuild and calibrated camera tracking.
- Later Wi-Fi reachability failed; final operating state is unverified.

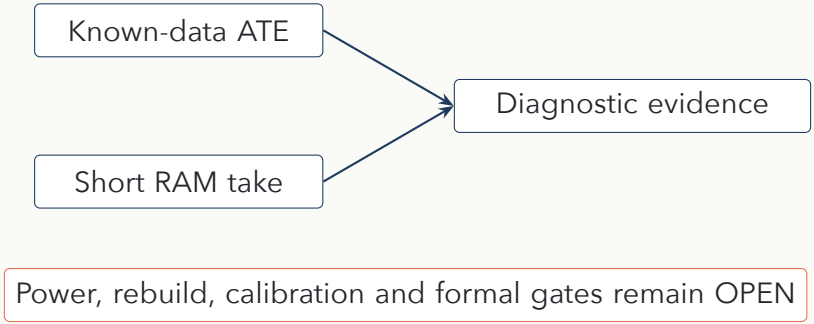


Figure 36: Useful observations coexist with unresolved qualification.

Neither larger RAM nor a clean short take establishes robot readiness. (Figure 36)

36. Next diagnostic sequence

PLANNED / diagnostic only

After explicit CAMERA AVAILABLE authorization, the addendum proposes this diagnostic sequence. It does not bypass formal N1 → N2 → N3 qualification.

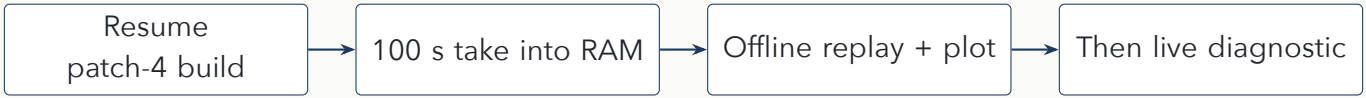


Figure 37: Proposed sequence, not executed in deck authoring.

Stable power and calibration remain prerequisites to qualification. (Figure 37)

37. Current system claim

MEASURED baseline / integration PLANNED

- The Pi build estimated motion on known EuRoC input with about 4 cm ATE.
- Camera recording improved in a short RAM test; camera estimation remains unverified.
- STM32H7 retains local balance and motor-safety ownership.

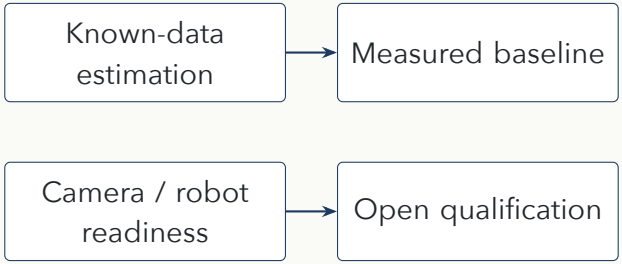


Figure 38: Progress and readiness answer different questions.

Resume the repair and diagnostic sequence while keeping every gate OPEN. (Figure 38)

References

[1] Campos, Elvira, Gómez Rodríguez, Montiel, Tardós. ORB-SLAM3. arXiv:2007.11898v2, 2021. Primary source.

[2] Burri et al. The EuRoC micro aerial vehicle datasets. International Journal of Robotics Research, 2016. Primary source.

[3] ROS 2 Jazzy: Quality of Service documentation. Primary source.

[4] ROS REP-103 / REP-105: units, coordinates and frames. Primary source.

[5] RealSense librealsense v2.58.4: timestamp reader and calibration parser. Primary source.

[6] ETH Zürich ASL: Kalibr camera–IMU calibration. Primary source.

[7] gaowenliang: imu_utils maintainer documentation. Primary source.

[8] Michael Grupp: evo trajectory metrics. Primary source.

[9] Raspberry Pi 5 / Active Cooler and RealSense D435i vendor records. Primary source. D435i record.

N4A accuracy is measured; D435i and robot qualification remain OPEN.

Backup: 4.2 N0: acceptance and diagnosis

PLANNED / no gate PASS

ARM64: 64-bit Arm; OS: operating system; UTC: Coordinated Universal Time.

PASS requirement	Measurement / clock
ARM64; Ubuntu 24.04; Jazzy	OS/package inventory; observation UTC
UTC synchronized; flags 0x0	Time service + raw power flags; current boot
Temperature $< 75\,^{\circ}\mathrm{C}$; available memory ≥ 512 MiB	Temperature + MemAvailable; sampled host time

FAIL: sticky undervoltage points to supply/cable; heat points to cooling; clock unsynchronized blocks cross-log correlation. Idle PASS does not prove loaded health.

These platform prerequisites are carried into N1 and checked again under load.

Backup: 4.3 N1: acceptance and diagnosis

PLANNED / no gate PASS

PASS requirement	Measurement / clock
Own direct branch $\geq$5000M	lsusb -t + camera sysfs speed; before/after UTC
Video; Human Interface Device (HID); Industrial Input/Output (IIO)	Access + owner inspection; no broad permissions
Identity; eligible sealed receipt	Host/boot match; age $\leq$2 h, no reboot

FAIL: 480M $\rightarrow$ cable/port/hubs; video works but motion denied $\rightarrow$ HID/IIO controls; serial mismatch $\rightarrow$ identity/descriptor audit, not firmware update.

Re-run N1 after storage qualification or a topology change.

Backup: 4.5 N2: delivery criteria

PLANNED / no gate PASS

PASS threshold	Measurement / clock
IR 14.25–15.75 Hz; gap $\leq$0.20 s	N/requested duration; interior rate; SDK stamp and arrival gaps
Gyro 190–210 Hz; gap $\leq$0.05 s	Whole-request / interior windows; per-stream stamps + arrivals
Accel 95–105 Hz; gap $\leq$0.05 s	Same method; do not compare device and host epochs
Positive stamps/frame steps; no missing IR frames	Adjacent differences in each stream; internal/trailing gaps strict

Burst-only delivery can pass first/last rate while failing whole-request rate.

Backup: 4.5 N2: content, health and acceptance

PLANNED / no gate PASS

PASS requirement	Measurement / clock
20 / 105 / 300 s stages complete	Requested duration + clean finalized shutdown; host interval
Coherent Y8; finite plausible motion	Dimensions, byte count, stats / hashes; sample epochs
Clean USB/kernel; stable platform	Continuous journal + one-Hz health; UTC/host times
Actual human review; warning retained	Explain startup offsets; retain calibration fallback verbatim

FAIL: torn images → USB/raw; motion-only failure → HID; late shutdown → controls.

Backup: 4.7 N3: PASS/FAIL matrix

PLANNED / no gate PASS

Transform (TF) messages describe coordinate relationships.

PASS requirement	Measurement / clock
60 s live + load delivery	IR/gyro/accel N2 bands; ROS header and host steady gaps incl. boundaries
Frames, profile, ten topics / types	Readback + image/info agreement + static TF; header stamps
Clean finalized bag / exact replay	Stored vs complete observed counts, order, header and decoded field hashes
No contamination / process fault	Live-domain guard + player/observer shutdown; host event time

FAIL: topic silence → driver/graph/QoS; missing context → durability; under-load gaps → storage.

Backup: 4.7 Storage prerequisite: no mount fallback

PLANNED / no gate PASS

Required threshold	Measurement / clock
Actual external device / UUID	findmnt, lsblk, bridge + physical identity; inspection UTC
$\geq$1 GiB and $\geq$60 s; both rates $\geq$9.216 MB/s	Overall and worst 64 MiB fsync batch; host interval
Matching SHA-256 readback; cleanup	Digest / cleanup result; 10 GiB headroom retained

The 8 GiB default may finish before 60 s on a fast device: incomplete duration, not low throughput. A larger write budget needs review; the IR1 margin does not qualify stereo bags.

FAIL: directory resolves to microSD $\rightarrow$ mount; slow batches $\rightarrow$ storage before ROS tuning.

Backup: 4.7 Content checks beyond delivery rates

PLANNED / no gate PASS

Runbook proposal	Measurement / clock
Stationary mean gyro norm $\leq$0.10 rad/s	Finite raw gyro vectors over declared stationary window; header epochs
Stationary mean accel norm 9.0–10.5 m/s^2	Finite raw accel; units/frame and stationary interval documented
Zero incomplete/corrupt notifications	Unfiltered SDK/kernel startup-to-stop log; UTC/run interval

These are native bring-up engineering criteria, not manufacturer specifications. Audit the prepared validator separately: matching rates do not prove all manual content criteria.

CameraInfo/TF and plausible stationary vectors do not prove calibrated camera–IMU timing.

Backup: 4.8 Stereo extension

PLANNED

- IR1-only N2/N3 remain preserved transport baselines. (Figure 39)
- Stereo adds reviewed IR2 profile, paired pixels and context.
- Requalify bandwidth, recording load, storage and calibration.
- Updated nodes, messages or derived formats need approval.

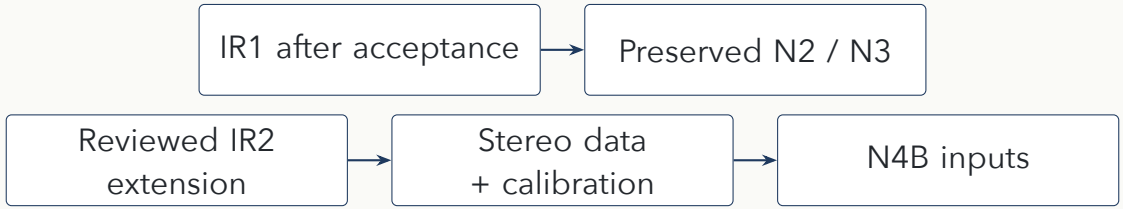


Figure 39: Stereo extension; schematic.

The stereo goal does not authorize quietly changing N2.

Backup: 4.9 N4A: estimator acceptance contract

PLANNED / no gate PASS

Absolute Trajectory Error (ATE) scores aligned paths; Relative Pose Error (RPE) scores motion increments.

Required / open threshold	Measurement / clock
Finite ordered trajectories; coverage	Export epochs in dataset time; coordinate/format audit
Repeat initialization / tracking	All attempts incl. failure; first eligible dataset epoch + host events
Accuracy / continuity budgets UNKNOWN	Reference association, rigid alignment, ATE/RPE; dataset epochs

FAIL: load/link error → build; empty images → reader paths; no inertial initialization → settings/time/motion. No accepted numeric budget means no performance PASS. N4A RPE is unverified in Phase-8 permitted sources; stable-power reruns and reviewed budgets remain missing.

MH01 accuracy is measured; N0 power health and this contract remain incomplete.

Backup: 4.10 N4B: input and run criteria

PLANNED / no gate PASS

Required / open threshold	Measurement / clock
Lossless stereo + valid pairing	Counts/pixels, epipolar residuals, left/right stamp difference; source domain
Calibrated geometry, noise and offset	Holdout residuals; camera/IMU units and common time relation
Repeatable finite tracking; budgets UNKNOWN	All trials, epochs, initialization/loss; no GT means accuracy UNKNOWN

FAIL: vertical mismatch → rectification; missing brackets → motion coverage; bias instability → clocks/calibration before feature tuning. Residual limits are not yet approved.

> Factory nominal geometry and assumed numeric noise do not qualify final VIO.

Backup: 4.10 N4B: immutable adapter provenance

MEASURED binary build / replay OPEN

Derived input contract

Preserve raw bags. Manifest links input, converter, settings and calibration hashes; record domains, offset sign, interpolation brackets and every rejected sample.



Figure 40: Frozen inputs make estimator comparisons attributable.

In Figure 40, test lossless pixels, exact nanoseconds, no extrapolation and count reconciliation.

Backup: 4.11 N4C: live PASS/FAIL criteria

PLANNED / no gate PASS

Required / open threshold	Measurement / clock
Valid paired input + IMU coverage	Source epochs; renewed stereo transport metrics
Initialization / continuity budgets UNKNOWN	Attempt count; loss/reset state; source epochs + host event times
Latency / queue bounds UNKNOWN	Queue, compute, receipt/output on host steady; mapped capture with uncertainty

FAIL: offline works, live lags → SDK callback/queue in the direct path; ROS delivery in the planned path; thermal stalls → N0 under load; loss in both → input/calibration first. Missing counters stay UNKNOWN.

Approve bounds before a live performance verdict; do not tune around unmeasured age.

Backup: 4.12 N5: measurement criteria

PLANNED / no gate PASS

Required / open threshold	Measurement / clock
ATE/RPE only with valid reference	Associated source epochs; rigid alignment; overlap and exclusions
P50/P95/P99 + initialization/loss	Named interval, estimator and sample count; host steady / source epochs
Resources, losses, queue depth	One-Hz health + event join; sequence/count reconciliation; unknown counters marked

All numeric performance budgets are UNKNOWN. FAIL: cross-domain subtraction → clocks; suspicious low ATE → alignment/association; averages hide stalls → raw timeline.

Continuity denominator: Count all initialization attempts; define first eligible input. Record state, reset/session ID, loss intervals and trajectory coverage.

Missing ground truth blocks accuracy claims; it does not force synthetic zero errors.

Backup: 2.6 QoS compatibility

KNOWN ROS semantics

- Quality of Service (QoS) is an offered/requested contract. [3]
- A reliable reader cannot match a best-effort-only writer.
- Sensor observers use best-effort, volatile subscriptions.
- Static transforms use reliable, transient-local delivery.

Writer	Reader	Compatible
Best effort	Reliable	No
Reliable	Best effort	Yes
Volatile	Transient local	No

> Topic names alone cannot prove a compatible connection.

Backup: 3.4 Topic and type contract

PLANNED / no gate PASS

Topic below /camera/camera	Message / cadence	Reader policy
`infra1/image_rect_raw`	`sensor_msgs/msg/Image`; 15 Hz	Best effort / volatile
`infra1/camera_info`	`sensor_msgs/msg/CameraInfo`; 15 Hz	Best effort / volatile
`gyro/sample`; `accel/sample`	`sensor_msgs/msg/Imu`; 200 / 100 Hz	Best effort / volatile
`/tf_static` (absolute topic)	`tf2_msgs/msg/TFMessage`; static	Reliable / transient local

Rates are requested baseline values; verify offers, depth and actual delivery.

Backup: 3.11 Robot transform ownership

PLANNED / no gate PASS

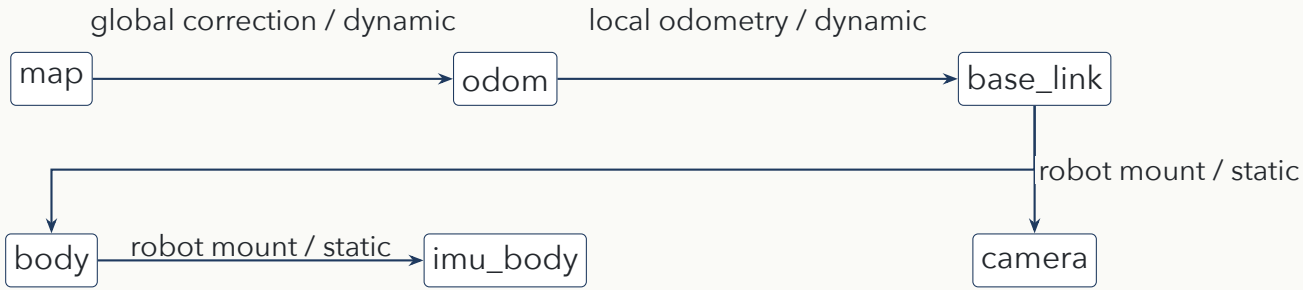


Figure 41: Dynamic edges need separate planned estimator and continuity owners.

Robot description owns static mounts, including base_link → body. The planned bridge owns dynamic edges; implementation is UNKNOWN. Optional world → map needs an explicit external-reference owner [4].

In Figure 41, one parent and one publisher per child; never relabel a jumping map pose as odom.

Backup: 3.12 Camera transform ownership

PLANNED / no gate PASS

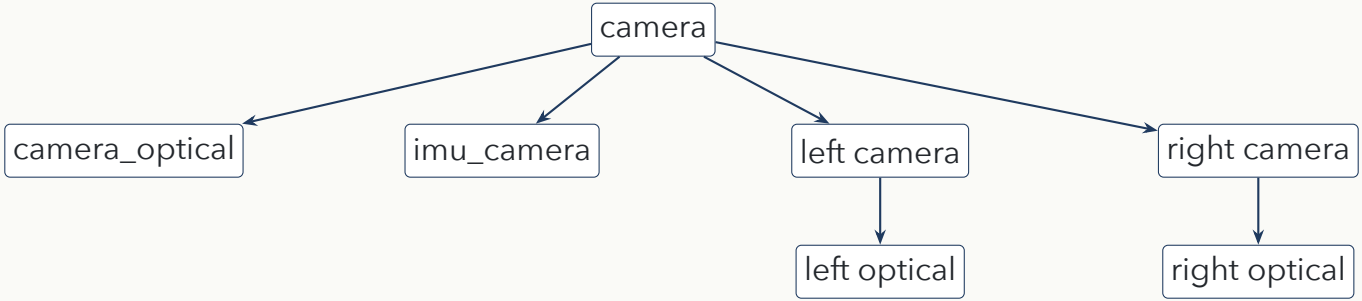


Figure 42: Wrapper-owned sensor geometry is static; frame names here are conceptual.

Wrapper /camera/camera publishes sensor edges on /tf_static when tf_publish_rate=0. Inspect actual parents and frame IDs; do not add this conceptual tree as a second publisher.

In Figure 42, the robot description owns the camera mount; the wrapper owns internal sensor geometry.

Backup: 3.16 Where timestamps are stamped

PLANNED / no gate PASS

Field / event	Stamp owner	Interpretation
t_{dev} / sensor frame	Device + SDK metadata	Hardware-relative or mapped domain; inspect each frame
t_{hdr} / message header	ROS wrapper	Producer epoch; not publish time
t_{rx}, t_s, t_e, t_o	Observer / adapter instrumentation	Local steady-clock receipt, start, end, emission
t_{bag} / storage record	rosbag2 / storage backend	Recorder time; never substitute for source header

A trajectory stamp is the estimated pose epoch, not when its file was written.

Backup: 3.17 Global time

PLANNED

- Global-time option readback is a configuration observation. [5]
- The SDK maps device time toward host time.
- Enabled global time does not measure camera–IMU offset.
- Arrival minus mapped stamp is not pure USB latency.

Observation	What it establishes
Option enabled	Requested timestamp behavior
Frame domain	Per-frame clock classification
Mapped timestamp	Relationship with mapping uncertainty
Camera–IMU offset	Separate physical alignment measurement

> Record domains and uncertainty before reporting source age.

Backup: 3.18 Worked latency decomposition

PLANNED / no gate PASS

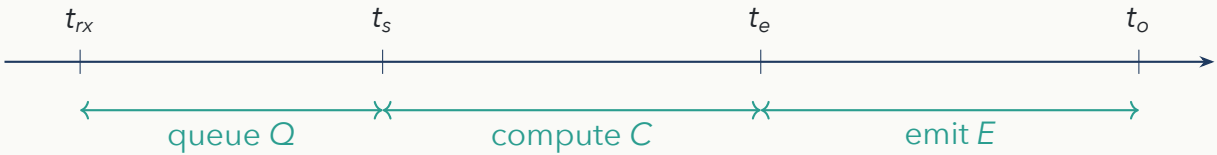


Figure 43: All three components use the same host steady clock.

$L_{rx \to o} = (t_s - t_{rx}) + (t_e - t_s) + (t_o - t_e) = Q + C + E.$

Capture age requires $\widehat{t}_c = g(t_{dev})$ in this domain. If mapping error is bounded by u, true age lies in $[t_o - \widehat{t}_c - u,\ t_o - \widehat{t}_c + u]$.

In Figure 43, fast compute does not imply fresh output when queue wait dominates.

Backup: 3.22 IMU noise

PLANNED

- Allan deviation compares noise across averaging time τ. [7] (Figure 44)
- Short-time slope helps estimate white-noise density.
- Longer records are needed to constrain bias instability or walk.
- Vendor comma-separated values (CSV) round motion; use full precision.

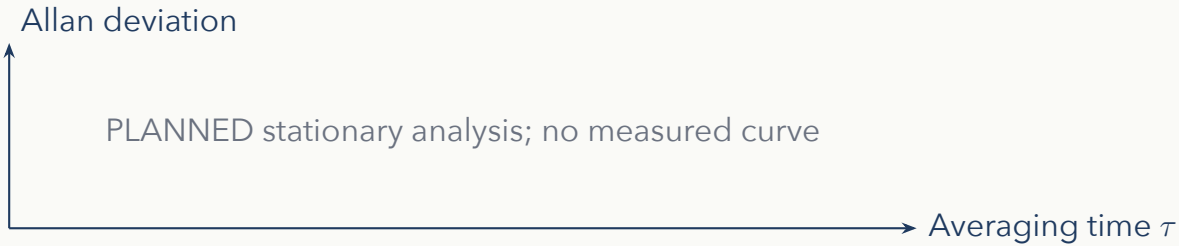


Figure 44: IMU noise; schematic.

Choose stationary duration from the averaging times you need.

Backup: 3.28 Settings audit before tuning

PLANNED / no gate PASS

Pinned parser group	Meaning / check	Failure implication
Camera1 / Camera2; Stereo	Projection, distortion, b or T_c1_c2; selected rectification path	Wrong disparity geometry; scale / epipolar error
IMU.T_b_c1	Camera → estimator IMU; numeric basis-vector / inverse test	Axis/sign error; gravity or bias inconsistency
IMU.Frequency; four noise fields	Continuous-time densities / walks; discretize once	False confidence; unstable initialization
ORBextractor fields	Feature count, pyramid, FAST thresholds	Tune only after input, timing and calibration

The active stereo candidate uses nominal geometry and assumed noise; qualification remains OPEN.

Backup: 3.28 Numeric IMU settings contract

PLANNED / no gate PASS

YAML (YAML Ain't Markup Language) settings must follow the selected pinned parser. Unit-specific noise is UNKNOWN; the diagnostic candidate uses ASSUMED upstream numeric values.

Key family	Interpretation
IMU.NoiseGyro / NoiseAcc	Continuous-time gyro / accel noise densities; code multiplies by $\sqrt{f}$
IMU.GyroWalk / AccWalk	Bias random-walk quantities; code divides by $\sqrt{f}$
IMU.Frequency / T_b_c1	Input-grid frequency f / camera-to-IMU transform; not raw accel rate

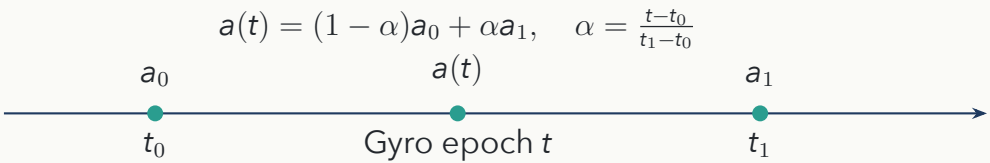
Camera.width/height/fps and Camera1/Camera2 keys must match actual inputs. Example EuRoC values describe that dataset; they are not D435i calibration.

> Do not discretize densities twice; assumed upstream values enable diagnostics, not qualification.

Backup: 3.29 Input association

PREPARED helper / runtime association UNVERIFIED

- Prepared helper interpolates accel only between valid brackets. (Figure 45)
- Use gyro epochs for a reviewed combined input grid.
- Reject gaps, duplicate/backward time and extrapolation.
- Retain originals and every interpolation/rejection decision.



Reject without valid brackets; never extrapolate.

Figure 45: Input association; schematic.

A 200 Hz input grid does not make raw accel 200 Hz.

Backup: 5.2 ATE computation outline

PLANNED / no gate PASS

Absolute Trajectory Error (ATE) compares associated poses after a declared alignment. P_i is the estimate and Q_i the reference; root mean square error (RMSE) summarizes translation [8].

$$E_i = Q_i^{-1} S P_i, \qquad e_i = \| \operatorname{trans}(E_i) \|, \qquad \mathrm{RMSE} = \sqrt{\tfrac{1}{N} \textstyle\sum_i e_i^2}$$

Figure 46: Rigid alignment S removes reference-origin choice, not a stereo scale error.

Associate epochs within a predeclared tolerance; retain unmatched samples. Convert both trajectories to the same pose direction and units before fitting S. Report valid overlap and failure trials.

In Figure 46, without independent reference data, D435i absolute accuracy stays UNKNOWN.

Backup: 5.3 RPE and pair policy

RPE UNVERIFIED in permitted sources / no gate PASS

Relative Pose Error (RPE) compares estimated motion with reference motion over a declared interval [8].

$$E_{ij} = (Q_i^{-1} Q_j)^{-1} (P_i^{-1} P_j)$$

Translation and rotation are separate metrics.

Figure 47: Relative deltas expose local drift rather than global alignment choice.

Pair policy	Question answered
Fixed time separation	How wrong is motion over that duration?
Fixed distance separation	How does error grow per traveled distance?
Consecutive vs all eligible pairs	Different weighting; report choice and pair count

In Figure 47, never compare RPE numbers with different delta units or pairing policies.

Backup: 5.6 Latency and loss

PLANNED

- P50/P95/P99 are the 50th/95th/99th latency percentiles.
- Name receipt-to-start, processing and receipt-to-output separately.
- Reconcile sequence gaps, received counts and recorded counts.
- Unavailable queue/drop counters remain UNKNOWN.

Metric	Required evidence
Latency percentiles	Named interval / clock / count
Loss	Sequence and count reconciliation
Queue depth / drops	Measured counter or UNKNOWN

> Percentiles need event definitions and eligible sample counts.

Backup: 5.7 Resource measurements

PLANNED

- Log Central Processing Unit (CPU) load and clock frequency. (Figure 48)
- Track temperature, memory and Resident Set Size (RSS).
- Record throttling, queue depth and free data space.
- Align samples with initialization, loss and optimization events.

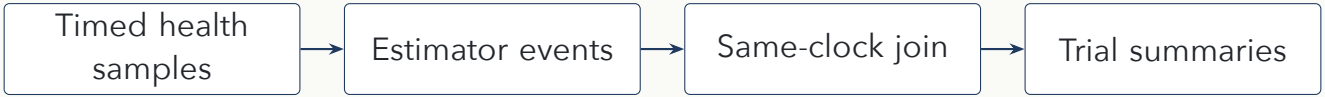


Figure 48: Resource measurements; schematic.

Resource averages cannot explain tail stalls alone.

Backup: 5.9 Timing results

MEASURED wall durations / timing qualification OPEN

N4A wall durations: 301 / 282 / 284 s for 184 s of dataset time. These include vocabulary load; live under-voltage invalidates a speed verdict.

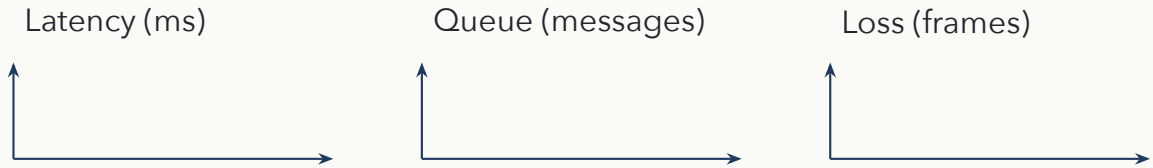


Figure 49: Wall duration is observed; per-frame latency, queues and loss remain unmeasured.

Do not convert wall duration into P50/P95/P99 processing latency. Separate startup, steady tracking and shutdown on a stable-power rerun.

Offline completion and real-time processing answer different questions. (Figure 49)

Backup: 5.10 Resource results

MEASURED N4A summaries / missing series UNKNOWN

N4A observed summary	Value / interpretation
Peak process RSS, runs 1 / 2 / 3	823 / 828 / 828 MiB
Live under-voltage / throttling	87 / 178 five-second samples = 49%
Maximum sampled temperature	63 °C; not a thermal time series

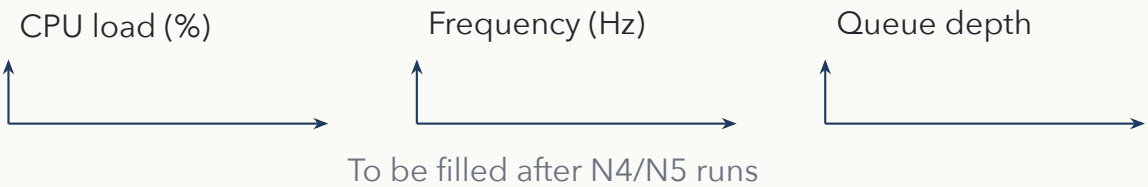


Figure 50: Summaries have evidence; missing joined time-series panels stay empty.

Stable power is required before interpreting processor performance. (Figure 50)

Backup: 3.32 Planned Pi-to-STM32 boundary

PLANNED / no gate PASS

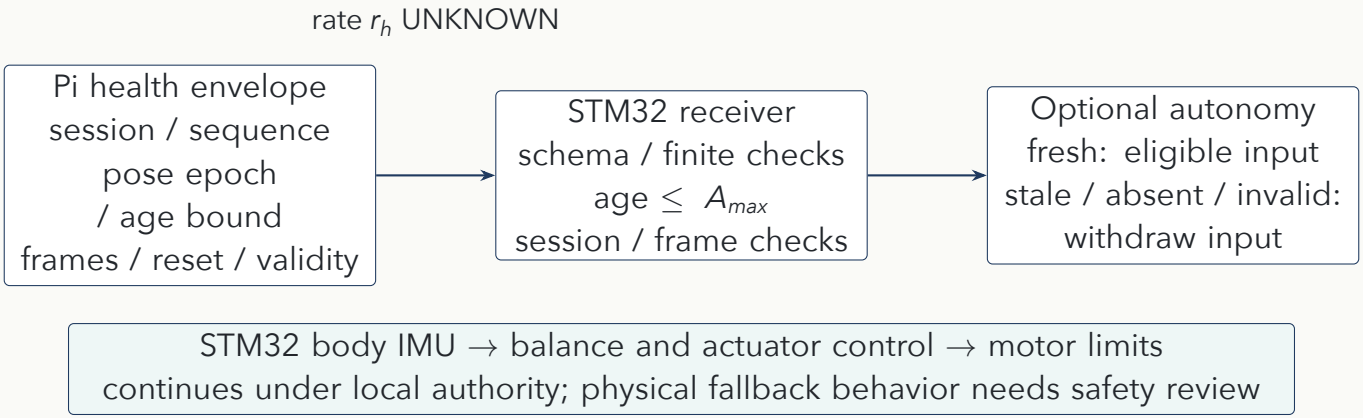


Figure 51: The health path provides optional information; local control keeps authority.

A_{max}, transport and r_h require approval. Receiver age must include source age and link residence with a defined clock relation; receipt-only timeout is insufficient.

In Figure 51, these are planned semantics; no message, firmware or stability test is implemented.

Backup: 4.13 N6: planned acceptance table

PLANNED / no gate PASS

PASS contract / open threshold	Measurement / clock
Reject every declared invalid fixture	Compare expected/actual state + reason; synthetic event time
Age bound $\leq A_{max}$; value UNKNOWN	Source-age bound + receiver elapsed time; defined Pi/STM32 relation
Reset re-establishment explicit	Session/frame sequence; no silent recovery

FAIL: old pose accepted → source-age semantics; wrap breaks timeout → receiver clock; old session accepted → reset protocol. Rate and timeout remain unselected.

> Implement only after Application Programming Interface (API) and message review; do not infer acceptance from a heartbeat.

Backup: 4.14 N7: planned acceptance table

PLANNED / no gate PASS

MCU denotes the microcontroller unit; its local clock measures controller actions.

Required / open threshold	Measurement / clock
No perception authority over safety	Ownership / code review + controlled fault trace
Local control retained on Pi loss	STM32 loop/watchdog/motor state; MCU clock
Fallback inside approved safety budget	Fault onset to local action; MCU domain; numeric bound UNKNOWN

FAIL: balance depends on ROS heartbeat → ownership; stale reference retained → fallback semantics; link reset bypasses validation → session handling.

No approved setup and bound means no physical integration PASS.

Backup: 1.3 Objectives and research questions

PLANNED

- Qualify physical input before estimation.
- Reproduce estimation on immutable recorded inputs.
- Measure accuracy, continuity, latency and resource limits.
- Review a health boundary before controller integration.

Operational questions: Sustain exact input? Initialize repeatedly? Reject stale/reset poses? Identify the first resource limit.

> Each objective needs its own evidence and acceptance review.

Backup: 3.1 Machine ownership

KNOWN ownership

- Mac checkout is canonical; preserve its dirty work. (Figure 52)
- The Pi acquires and records locally.
- Secure Shell (SSH) carries control and small evidence.
- Use a separate checksummed runtime bundle for each revision.



Figure 52: Machine ownership; schematic.

Runtime identity: Hash dirty sources, binaries, libraries and configuration; retain backend, wrapper and launch readback.

> Deployment copies tools; it does not reset either checkout.

Backup: 6.1 Runtime and USB

PLANNED

- Default: native Pi baseline and direct known-good USB3 cable.
- Docker adds package, device and user/group boundaries.
- A hub/storage topology can change shared power and bandwidth.
- Revisit only after equivalent physical gates and clear deployment benefit.

Default	Evidence that changes it
Native Pi	Equivalent gates + packaging benefit
Direct USB3	Alternative topology sustains speed/power/load

> Packaging cannot compensate for a 480M camera branch.

Backup: 6.2 Inputs and clocks

PLANNED

- Keep exact IR1 640×480/15 qualification unchanged.
- N4 development selects IR1+IR2 Y8 640×480/30, gyro/200, accel/100.
- Development emitter is OFF for fixed-scene feature tracking.
- Preserve global-time readback and original timestamp domains.

Default	Evidence that changes it
N2 IR1/15; N4 stereo/30	Independent delivery and load evidence
N4 emitter OFF	Reviewed controlled scene/lighting comparison
Logged global-time state	Measured domains/drift/offset evidence

Change one input variable with paired quality and load evidence.

Backup: 6.3 Algorithm and calibration

PLANNED

- Validate upstream EuRoC and offline inputs before live wrapper.
- Candidate: Kalibr spatial/time calibration plus measured IMU noise.
- Choose tools from import/build/model compatibility and residuals.
- Defer wheel odometry until timing, frames and slip evidence exist.

Default	Evidence that changes it
Upstream then offline	Matched-data wrapper equivalence
Measured calibration/noise	Tool compatibility + holdout residuals
Defer wheel fusion	Synchronized geometry + slip evidence

Tool qualification: Kalibr spatial/time calibration and imu_utils noise analysis are candidates. Test bag import, models and holdout residuals.

> Add complexity only to solve a measured failure mode.

Backup: 7.1 Historical evidence

HISTORICAL / MEASURED saved observations

- 8 September: x86 virtual machine (VM)/Docker USB2 NO-GO. (Figure 53)
- 12/13 September: Pi saw 5000M, then USB regressed to 480M.
- A historical 105-second capture did not complete N2.
- Archived optimistic PASS wording cannot promote current gates.

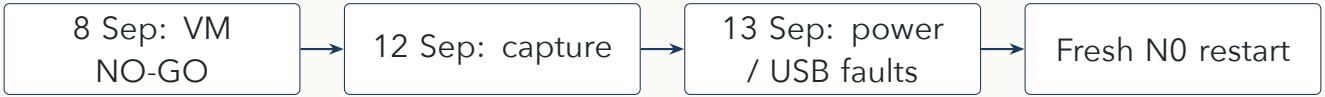


Figure 53: Historical evidence; schematic.

History explains risks; restart acceptance from fresh N0.